

Kettlebell training

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Introduction

The kettlebell (KB) is a cast-iron or cast-steel weight shaped like a cannonball with a handle on top (Figure 42.1). The name kettlebell stems from its distinctive form resembling an old water kettle. Originally developed in the 1700s as a counterweight for market scales in Russia, KBs have been used as training tools in eastern Europe since then (Tsatsouline, 2006). In the past decade, training with KBs has become increasingly popular in the worldwide functional training and fitness community. Since the 1960s, there have been competitive events, where athletes compete in different lifting forms. However, the competitive nature of KB sport will not be the focus of this chapter.



Figure 42.1 A selection of kettlebells with different weights and forms

What makes KB training unique and different from other forms of weight training? Unlike traditional dumbbells, the center of mass in a KB lies outside the hand, which predisposes KBs for ballistic and swinging movements involving the whole body (Jay et al., 2011). That being said, KB training usually consists of multi-joint movements and exercises focusing on functional or myofascial chains rather than on isolated muscles or body parts. Of course, this holistic approach to exercise and physical training is not necessarily limited to KB training alone. Other training forms such as Olympic lifting, powerlifting or so-called functional training also focus on complex movements rather than isolating muscles as in traditional bodybuilding. However, the swinging component of most KB exercises is indeed a unique feature that has potential benefits for the connective tissue network.

Traditionally, training with KBs has been popular in policemen, special forces and military personnel because it is thought to improve explosive power, strength endurance as well as mental toughness (Tsatsouline, 2006; Sukopp, 2016; Kwella, 2017). Today, its use has spread to fitness clubs and rehabilitation clinics around the world. Research has shown that KB training leads to acute responses in hormones triggering muscle adaptation (Budnar et al., 2014), reduced neck and lower back pain (Jay et al., 2011), improved postural stability (Jay et al., 2013), increased aerobic fitness (Thomas et al., 2014;

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Falatic et al., 2015) as well as enhanced maximal and explosive strength (Lake and Lauder, 2012A). Therefore, it serves as a good addition to any regular health and fitness training regimen. The aim of this chapter is to elaborate how KB training may affect the fascial system and how it can be integrated into a training program tailored to improve connective tissue properties. One of the most commonly used KB exercises, the KB swing, as well as other exercises that can be utilized to impact the myofascial system, will be introduced and thoroughly described. After practicing the described examples, the reader will be able to apply the underlying principles to other KB exercises or similar forms of training.

Potential effects of KB training on the fascial system

To date, there is no scientific evidence on the impact of KB training on the fascial system. However, the swinging movements in many KB exercises could affect connective tissue renewal as well as its energy storage capacity. Mechanical loading, especially eccentric muscle work, has been shown to induce increased collagen remodeling in tendons (Kjær et al., 2009). As swinging the KB often requires an eccentric deceleration at the point of movement reversal, such an effect may also occur in KB training. In this context, it has to be underlined that the three-dimensional nature of the exercises may stimulate the fibers of the fascial system more effectively than movements occurring in one plane only. The storage and release of elastic energy is one of the main tasks of our connective tissue during functional tasks such as walking (Ishikawa et al., 2005). The KB might be an ideal tool to train and improve this energy storage capacity, because smooth movement execution and swinging exercises as the ones used in KB training have generally been

proposed as positive training stimuli for the fascial system (Schleip and Müller, 2013).

The KB swing

The KB swing unarguably represents one of the most commonly used KB exercises. It is an exercise with relatively low technical demands that can be executed with relatively light loads. Both features make it an exercise that can easily be integrated into a training program for both novices and experienced lifters (Lake and Lauder, 2012B). Furthermore, its complex and ballistic nature involves most muscles of the posterior body (the hamstrings, glutes, erector spinae and latissimus dorsi) (Budnar et al., 2014) and the corresponding connective tissue, in particular the thoracolumbar fascia (TLF).

In the starting position, the feet are placed about a shoulder width apart, with the toes facing slightly outward. The arms should remain extended during the whole exercise, allowing the entire energy created by the hip, knee and core muscles to be fully transferred into the KB. The KB is placed on the floor, 5 cm in front of the feet. The movement is initiated by actively flexing, or “hinging” the hip joint with little to no flexion of the knee and a neutral spine. The KB is then actively pulled backwards between the knees and behind the pelvis (the “backswing”). The end position is governed by hamstring flexibility and the ability to maintain a neutral spine.

From this position, the hip is explosively extended to accelerate the KB forward and upwards. The forearms should remain in contact with the body during the swinging phase until the KB passes the hip, ensuring that the energy is fully transferred from the feet through the torso and arms into the KB. The knees, hips and upper body should all be fully extended while swinging the KB to about shoulder height. In the next

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step, either actively pull the KB back down or let gravity take over and wait until the KB is accelerated downwards. When the forearms touch the body, actively flex the hip. In this portion of the movement, the hamstrings, the glutes and the lumbar extensors work eccentrically, and the momentum and energy are absorbed in the connective tissue, like in an elastic spring. When the KB reaches the end position, the hips are again explosively extended to accelerate the KB forward and upward in a fluid movement. During the entire exercise, the movement should be led by the hip joint, not the arms and shoulders. Beginners tend to pull the KB cranially with their shoulders instead of using the momentum generated by the hip to swing the KB. One consequence is that they reduce weight because their shoulders fatigue rapidly. Also, the amount of force generated during hip extension will determine the height of the KB. If you cannot swing the KB high enough, do not use your arms but try to extend the hip faster after the backswing. See Figure 40.2 for a detailed illustration of the KB swing.

Key points for the KB swing:

- Actively pull the KB back to start the exercise
- Explosively extend hip and knees while keeping arms straight
- Swing KB to shoulder height, keeping the entire body extended
- Do not pull the KB upwards with your shoulders
- Keep the hips extended until the forearm touches your torso
- Flex hips (and knees slightly) to slow down the KB, absorbing the energy
- The movement should look smooth and easy

As mentioned, the KB swing mostly works on the muscles of the posterior body, especially the hamstrings, gluteal muscles and the latissimus dorsi, all of which are linked by connective tissue that has the ability to transfer force (Krause et al., 2016; Wilke et al., 2016) (Chapter 13). In both end positions, elastic energy can be stored during the eccentric movement phase in the myofascial system and released in the subsequent

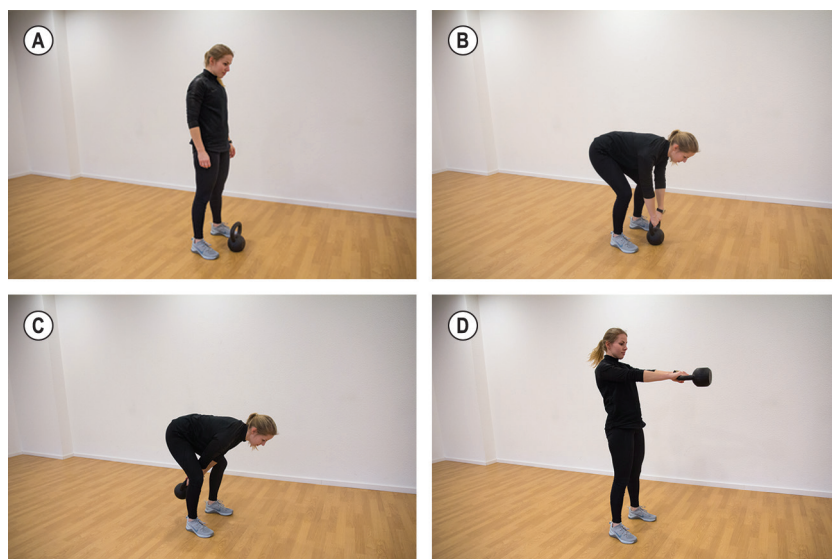


Figure 42.2 Detailed illustration of the KB swing

(A) Starting position. (B) Bottom position before the initiation of the "backswing". (C) End position of the "backswing". (D) Top position right in the turning point of the KB.

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concentric movement: in the flexed low position, the TLF as well as the fascia of the anterior shoulder and arm act like a spring, being stretched and releasing kinetic energy as the KB is accelerated in the anterior direction. In the extended high position, energy is stored and released in the myofascial tissue of the anterior body, as well as in the posterior shoulder and arm.

Alterations in TLF morphology (Langevin et al., 2009) and sliding properties (Langevin et al., 2011) have been shown in patients with low back pain. Similarly, changes in fascial thickness of the sternocleidomastoid muscle seem to manifest in patients with neck pain (Stecco et al., 2014). KB swing training might impact connective tissue remodeling and elastic properties of the TLF and the entire myofascial tissue of the shoulder-arm complex. These alterations may, in turn, explain the positive effect of KB exercises for patients with low back and neck pain (Jay et al., 2011). In sum, the KB swing seems to be a good exercise for preventing non-specific myofascial pain, as well as for treating patients with low back or neck pain, as long as the weight is chosen according to the functional abilities and the physical status of the patient.

One-handed KB swing

If one is able to perform a KB swing with proper form and good technique, progression can be achieved via the one-handed swing. Due to the unilateral character of the exercise, the core muscles are engaged in an anti-rotational manner, increasing complexity and effectiveness (Kwella, 2017). The stress on the muscles and the connective tissue on the KB swinging arm is also greater than in the two-handed swing. Additionally, one-handed swings are the foundation for more complex KB lifts like the KB clean or the KB snatch, and should therefore be mastered before further progressing the training routine.

The movement itself, as well as the general recommendations, are the same as for the two-handed swing. With regard to the grip, there are essentially three different ways to hold the KB, each of which has an impact upon muscle activation in the forearm and shoulder. The neutral grip is the most commonly used, although internal or external rotation of the shoulder and rotation of the KB by 90° to grab the corner of the KB in the starting position are also possible.

The “backswing” is again initiated by actively pulling the KB between the legs, backwards. Then the hip and knees are explosively extended while keeping the arm straight, swinging the KB to about shoulder height. On the return, ensure the weight and the momentum of the KB do not “twist” the body by actively engaging the muscles of the core (especially the external and internal obliques), to counteract the rotational forces which the KB creates while gravity accelerates it downwards. Keeping the forearm as close to the inside of the ipsilateral leg as possible, flex the hip when the KB reaches the midline, absorbing the energy of the KB and transferring it to the next hip extension movement to accelerate the KB back upwards (see Figure 42.3 for a detailed illustration of the one-handed KB swing.)

Key points for the one-handed swing:

- Actively pull the KB back with one hand to start the exercise
- Explosively extend hip and knees while keeping the arm straight
- Swing KB to shoulder height, ensuring the torso does not twist
- While the KB is accelerated downwards, make sure that the body stays straight
- The movement should look smooth and easy

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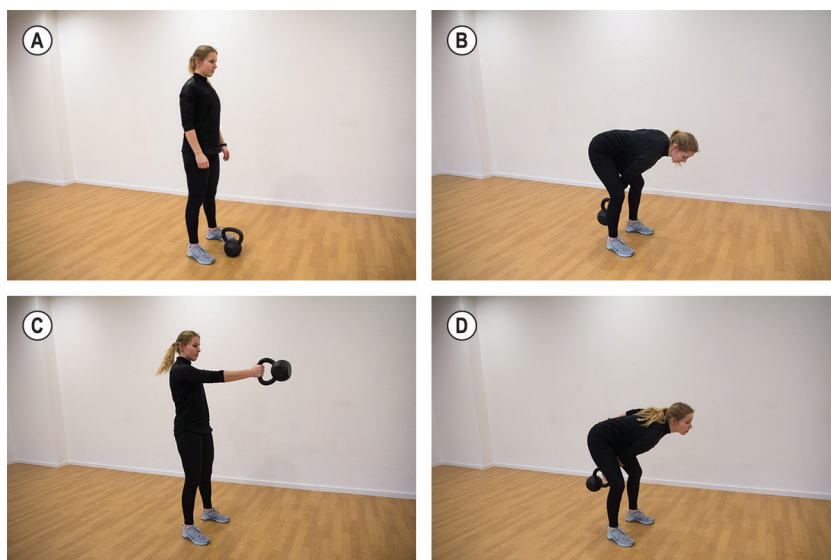


Figure 42.3 Detailed illustration of the one-handed KB swing

(A) Starting position. (B) End position of the "backswing". (C) Top position right in the turning point of the KB. (D) End position of the subsequent "backswing" phase.

Rotational outside swing

To focus even more on the three-dimensional structure of the connective tissue network of the body, a rotational component can be added to the conventional KB swing. The rotational outside swing creates the opportunity to play with the amplitude of both swinging height as well as rotation, thus targeting even more of the connective tissue's fibers and muscles. In the starting position, the KB lies on the floor next to the right or left foot (depending on which side the exercise is started). To grab the KB, internally rotate the contralateral femur and flex the hip, lift the KB off the floor with both hands, slightly swinging it forward as the hips are extended. While swinging the KB, rotate the hip back into a neutral position (facing forward) to create a straight standing position when the KB reaches shoulder height. While the KB swings to the contralateral side, internally rotate the other hip and slightly flex the hips and knees to decelerate the KB. Repeat the procedure to first decelerate and then accelerate the KB back to the starting position

(see Figure 42.4 for a detailed illustration of the rotational outside swing).

Key points for the rotational outside swing:

- Start with the KB on the floor next to one foot
- Internally rotate the contralateral hip to grab the KB
- Swing the KB forward and then to the opposite side while internally rotating the other hip
- Extend the hips back to neutral to accelerate the KB back forward and to the opposite side
- The movement should look smooth and easy

The rotational component of the exercise means the elastic energy in the bottom position is stored more in the glutes, the latissimus and the connecting TLF in a three-dimensional manner. When adjusting the amount of rotation, the exercise gives the perfect opportunity to reach more connective tissue fibers than with the normal two-handed KB swing. It is also possible to switch between the rotational out-

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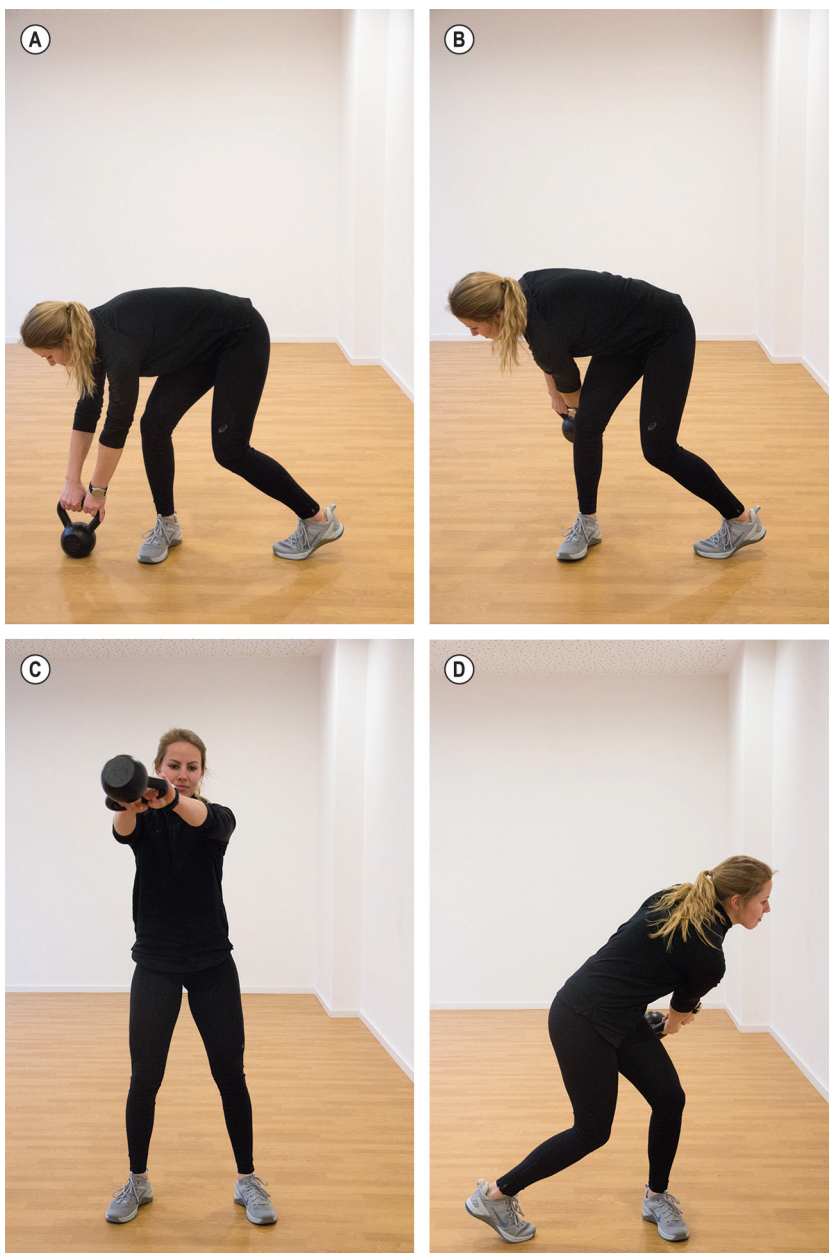


Figure 42.4 Detailed illustration of the rotational outside swing

(A) Bottom position before the initiation of the "backswing" (notice the internal rotation in the left hip joint). (B) End position of the "backswing". (C) Top position right in the turning point of the KB (notice the neutral position in both hip joints). (D) End position of the backswing at the contralateral side (notice the internal rotation in the right hip joint).

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side swing, the conventional KB swing and the one-handed KB swing, making this a complex workout that involves several functional and myofascial chains in one exercise.

Practical tips

What is the correct weight to start KB training as a novice?

Although it is not easy to give general recommendations, there is no need to start with less than 8 to 12 kg if you are healthy, depending on the exercise. For a woman, consider beginning with 12 kg, although a trained woman could start with 16–20 kg. For a man, start with 16–20 or 20–24 kg if a more experienced lifter (Kwella, 2017). If unsure, start with a lighter KB and then progressively increase the weight. Loading can be increased without increasing the weight by swinging the KB faster.

How many repetitions and sets should I start with?

As a beginner, it is important that you learn the exercise with proper form and technique. In KB training, the loading is usually determined using work and rest time in an interval training manner rather than counting repetitions as in traditional weight training. In the studies by Jay et al. (2011, 2013), a total of 10 sets with a ratio of 30 s of work and 60 s of rest was used in the first 4 weeks, which might be a good starting point for a beginner. In the last 4 weeks of their studies, the rest period was reduced to 30 s. Lake and Lauder (2012A) used 12 sets of 30 s with 30 s of rest during their 6-week study. However, you will find a plethora of interval training methods for KB training, each of which can be used effectively, depending on your training goal.

How should I warm up prior to KB training?

It is generally advised that a warm-up consists of a period of aerobic exercise, followed by dynamic stretching, and ending with a period of movement similar to the upcoming activity (McGowan et al., 2015). As you use almost the entire body during KB training, the dynamic stretching portion of the warm-up should also incorporate multi-joint exercises. As an even more specific warm-up, you can do the KB swing without weights, or with lower weights than you use during the actual workout. This approach has also been used in research (Jay et al., 2011, 2013) and seems an appropriate choice.

What other equipment is required to start KB training?

Besides a set of KBs with different weights, there is no need for additional equipment. Shoes with a hard sole are recommended; also, exercises can be performed barefoot, although there is the danger of dropping a KB and causing injury.

Conclusion

Although there is no scientific evidence for the impact of KB training on the fascial system, the swinging nature of KB exercises can potentially have a positive effect on connective tissue remodeling and energy storage capacity. Training with KBs has been shown to reduce back and neck pain. Although this is more of a speculative nature, it might partly be explained by normalization of connective tissue morphology and sliding properties due to KB swing training. Training with KBs focuses on entire myofascial chains and complex movements rather than on isolated muscles, which makes it a valuable addition to any exercise program. There are many more KB exercises than those described in this chapter, which is intended as a starting point to discover the benefits of KB as a training tool.

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